

The **BLE-Based Employee Tracking System** offers a range of intelligent features that make it efficient, user-friendly, and highly suitable for modern workforce environments. The system combines IoT technologies like BLE and ESP32 with cloud computing via Firebase, enabling seamless and automated employee tracking.

1. Contactless Attendance Logging

The system enables employees to be automatically logged in and out as they enter or exit the work premises. This reduces the need for physical attendance systems such as biometric scanners or manual registers.

2. Real-Time Detection Using BLE Beacons

Employees carry a BLE beacon that emits a unique signal. These beacons are detected in real-time by the ESP32 receiver placed at the site's entry point, ensuring timely and accurate logging.

3. ESP32 Receiver with Wi-Fi Support

The ESP32 microcontroller, known for its low power consumption and dual connectivity (Wi-Fi and BLE), detects beacon signals and transmits the data to the cloud in real time.

4. Firebase Cloud Integration

All entry and exit logs are stored in Firebase, allowing for real-time updates, seamless synchronization, and secure data storage accessible from anywhere.

5. Automatic Work Hour Calculation

The system intelligently calculates the total active work duration per employee each day using the time difference between entry and exit timestamps.

6. Web Dashboard for Monitoring

A responsive and user-friendly web dashboard provides authorized personnel (admin/supervisors) access to attendance logs, active hours, and status of each employee.

7. Passive Tracking

Unlike RFID or biometric systems, employees don't need to remember to scan a card or press a button. Their presence is passively detected when in range (up to ~30 meters), ensuring seamless operation.